

Abstract

Embrace the Unknown: Russell's Paradox and The Halting Problem

What to do when a very natural assumption about sets leads to a contradiction? A disastrous revelation that threatened foundations of mathematics at the end of the 19th century, Russell's Paradox uncovers our naive assumption about what we can or cannot do with sets. In this talk, we inspect the Paradox and its Theory of Computation counterpart, The Halting Problem using a proof reminiscent to Cantor's diagonalization argument.